

## Home Assignment

AI-Powered Testing Mastery

Course: Gen AI Test Tool Developer [AI Agentic Workflows]

Topic: LangChain & LangGraph

Live Session Date: 21<sup>st</sup> March 2026

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Using the Langflow-

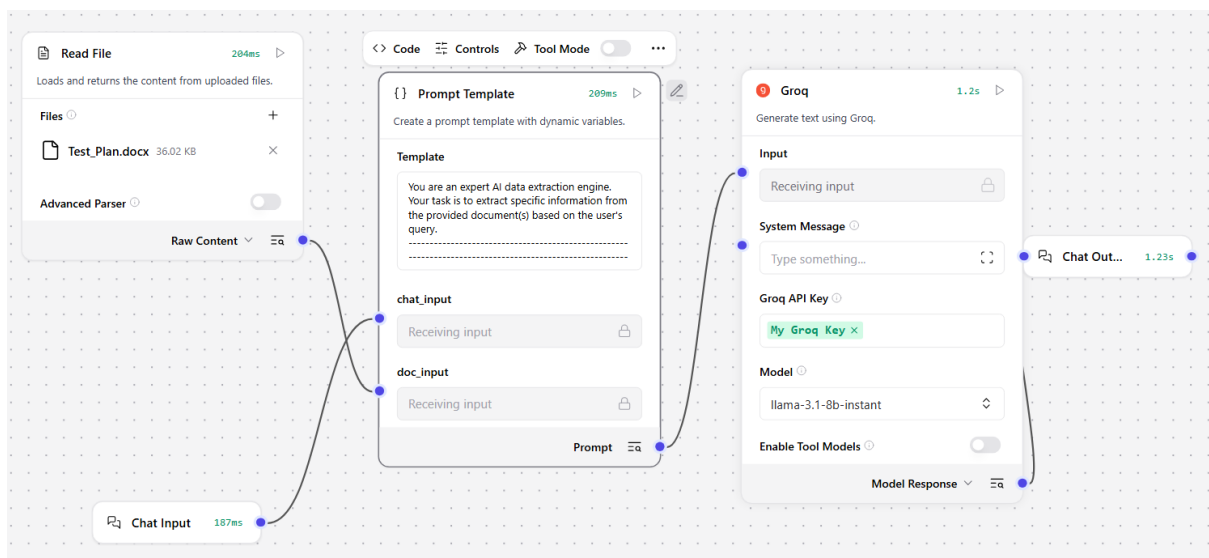
- **Develop a Workflow to read the document and extract the required data**
- We can add multiple documents.

### Building a Langflow RAG-Style Extraction Pipeline

#### “Document Reader & Data Extractor”

#### Components of the Workflow:

1. **Files** = Context (unstructured data source)
2. **Chat Input** = Query / Extraction instruction
3. **Prompt Template** = Orchestration logic (VERY critical)
  - The **Prompt Template** is the *core intelligence layer* here.
  - It determines how accurately your workflow extracts data.
4. **LLM (Groq)** = Reasoning + Extraction
5. **Output** = Structured result



For the 3<sup>rd</sup> Component '**Prompt Template**', the following prompt is used:

You are an expert AI data extraction engine.

Your task is to extract specific information from the provided document(s) based on the user's query.

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 **DOCUMENT CONTENT:**

{doc\_input}

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 **USER QUERY:**

{chat\_input}

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 **INSTRUCTIONS:**

1. Carefully read the user query and understand what data needs to be extracted.
2. Analyze the document content thoroughly.
3. Extract **ONLY** the relevant information that directly answers the query.
4. If multiple documents are provided, combine and reconcile the information.
5. If the answer is not found, respond with: "Data not found in the provided document(s)."

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 **OUTPUT FORMAT RULES:**

- Provide a clear, structured response
- Use bullet points or JSON format wherever applicable
- Do NOT include irrelevant text
- Do NOT hallucinate or assume missing data

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 **FINAL OUTPUT: Show ONLY the Output; NOT the above prompt**

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**User Queries:** To Validate the workflow, the following User Queries have been fed to the ChatBot:

1. Search Test Deliverables
2. Enlist only first 2 Test Deliverables

**LLM Response to the following Queries :** "Data not found in the provided document(s)."

3. What is your name?
4. Which LLM Model is used here
5. who won the T20 world cup?

